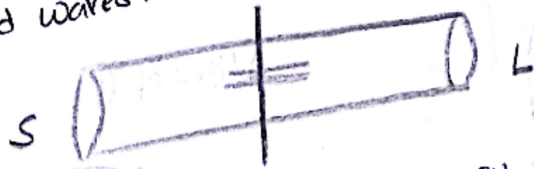


22. ELECTROMAGNETIC WAVES

i. String waves (Mech wave).

$$y = A \sin(\omega t - kx)$$

ii. Sound waves:



Air Particles Undergoes SHM.
In String Particles Undergoes SHM.

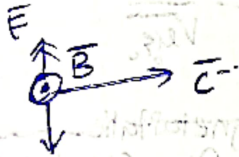
1. $y = A \sin(\omega t - kx)$ Displacement wave.

Value of Pressure is oscillating.

2. $\Delta P = (\Delta P)_m \cos(\omega t - kx)$ Pressure wave

E-M waves:

Electric, Magnetic Fields will oscillate.



The direction of Propagation of Electromagnetic wave is always in the direction of $\vec{E} \times \vec{B}$.

Cross Product of $\vec{E}$ & $\vec{B}$

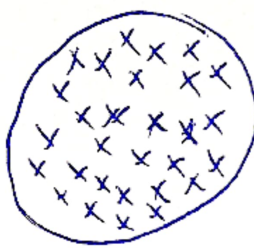
All the Electromagnetic waves are transverse waves.
Non Mech waves.

Displacement current:

1. Time Varying $\vec{B}$:

$$\frac{d\vec{B}}{dt}$$

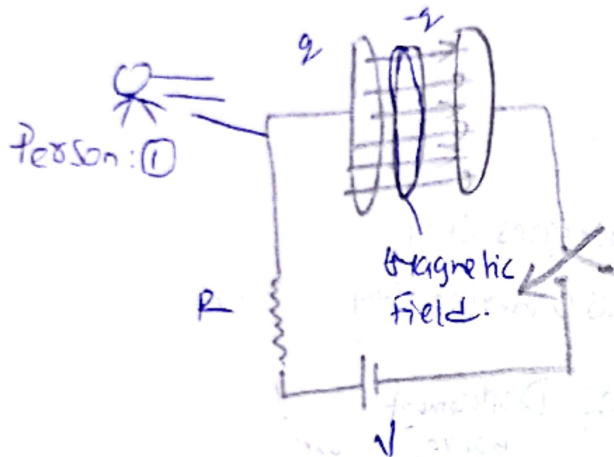
Time Varying $\vec{B}$ will produce $\vec{E}$...



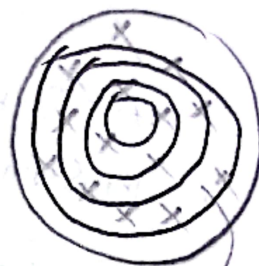
$$\oint_{\text{loop}} \vec{E} \cdot d\vec{l} = E_{\text{loop}} = -\frac{d\phi_B}{dt} \quad \sim \text{Faraday's Law}$$

2. Time Varying $\vec{E}$:

- Produces Magnetic field
- Maxwell $\rightarrow$ circular capacitor



View of ①



Displacement Current

$$E = \frac{\alpha}{\epsilon_0}$$

$$\oint_{\text{loop}} \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \left(\frac{d\phi_E}{dt} \right)$$

$\mu_0 \rightarrow$ Permeability

$\epsilon_0 \rightarrow$ Permittivity

$$C = \frac{1}{V \epsilon_0 \epsilon_r}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i_{\text{net}})$$

$$\oint \vec{E} \cdot d\vec{l} = \frac{q}{\epsilon_0}$$

closed

Magnetostatic - Amp Circuited Law.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 [i_d] \quad (\because i_d = \epsilon_0 \left(\frac{d\phi_E}{dt} \right))$$

$$F = qvB \sin \theta$$

Displacement Current.



Maxwell's Equations:

1. $\oint_{\text{closed}} \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} (q)$: Gauss's Law (Electricity)

2. $\oint_{\text{closed}} \vec{B} \cdot d\vec{s} = 0$: Gauss's Law (Magnetism)

3. $\oint_{\text{loop}} \vec{E} \cdot d\vec{l} = - \frac{d\phi_B}{dt}$: Faraday's Law.

4. $\oint_{\text{loop}} \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$: Amp Maxwell.

i. Speed of the Electromagnetic Wave: $c = \frac{E_0}{B_0}$

EM wave:

$E = E \sin(\omega t - kx)$

$E = E \sin\left[\omega\left(t - \frac{kx}{\omega}\right)\right]$

ii. $E = E \sin\left[\omega\left(t - \frac{x}{c}\right)\right]$

iii. $B = B \sin\left[\omega\left(t - \frac{x}{c}\right)\right]$

iv. $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ (Vaccum)

v. $c = \frac{1}{\sqrt{\mu \epsilon}}$ (Other Medium)

vi. Relative Permittivity:

$\epsilon_r = \frac{\epsilon}{\epsilon_0}$

$E_r = \frac{E}{E_0}$

Energy Density
of
EM waves

Intensity of EM
waves:

Momentum of EM
Wave.

$$\psi = \psi_E + \psi_B$$

$$\psi = \frac{1}{2} \epsilon_0 E^2 + \frac{B^2}{2\mu_0}$$

$$\psi = \frac{1}{2} \epsilon_0 E_0^2 \sin^2(\omega t - kx) + \frac{B_0^2 \sin^2(\omega t - kx)}{2\mu_0}$$

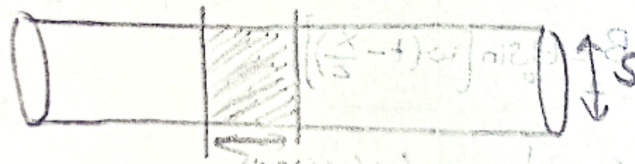
Avg Energy density:

$$\psi = \frac{1}{4} \epsilon_0 E_0^2 + \frac{B_0^2}{4\mu_0}$$

$$\psi = \frac{1}{2} \epsilon_0 E_0^2$$

$$\psi = \frac{B_0^2}{2\mu_0}$$

Intensity of EM waves:



The Energy transmitted by the wave Per Unit area Per Unit time

$$I = \frac{U}{\Delta t \cdot S} \quad \text{--- (1)}$$

$$\psi = \frac{\text{Energy}}{\text{Volume}} = \frac{U}{S \Delta x} \quad \text{--- (2)}$$

$$\frac{I}{\psi} = \frac{\Delta x}{\Delta t} = c$$

$$I = \psi c \quad \text{***}$$

Intensity = Energy Density x Speed of light.

$$I = \frac{1}{2} \epsilon_0 E_0^2 c = S_{avg}$$

$$E_{\text{photon}} = \frac{hc}{\lambda}$$

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

$$E_{\text{waves}} = \frac{E}{c}$$

Radiation Pressure:

$$R_p = \frac{I}{c} \quad \text{Perfectly Absorbing}$$

$$R_p = \frac{2I}{c} \quad \text{Perfectly Reflecting Constant...}$$

$$\epsilon = 8.85 \times 10^{-12}$$

$\epsilon_0 \rightarrow$ Permittivity
 $B_0 \rightarrow$ Magnetic field (Amplitude)
 $E_0 \rightarrow$ Electric field

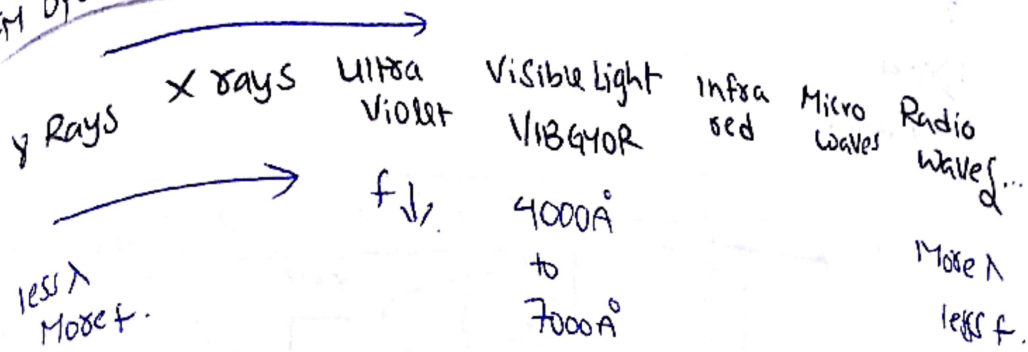
$$v = \frac{c}{\sqrt{\epsilon_r}}$$

Speed of light in any medium

Relative Permittivity $\epsilon_r = \frac{\epsilon}{\epsilon_0}$

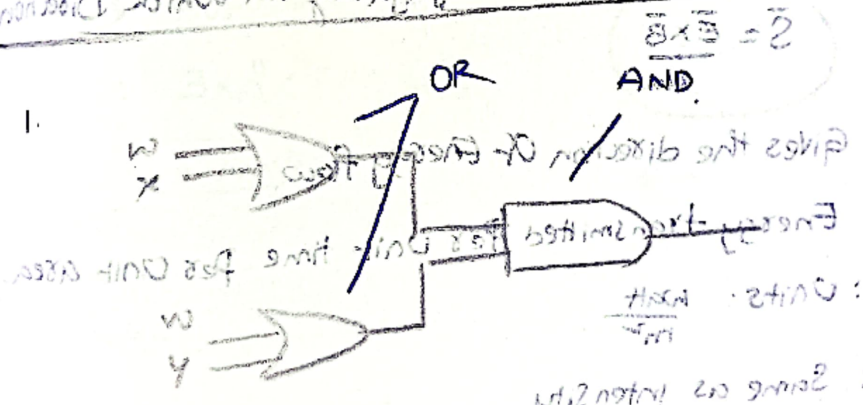
EM Spectrum :

λ increases.

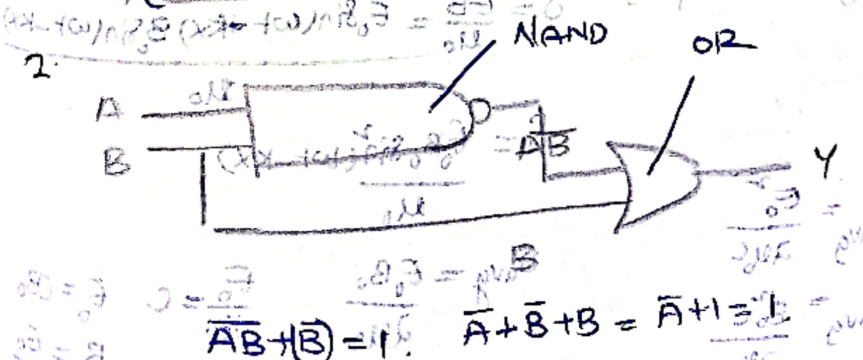


Boolean Algebra.

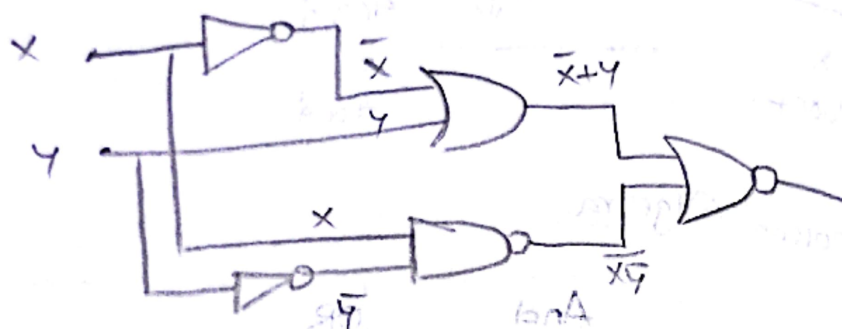
	And	OR
	$0A = 0$	$0+A = A$
	$1A = A$	$1+A = 1$
	$AA = A$	$A+A = A$
	$A\bar{A} = 0$	$A+\bar{A} = 1$
Commutative	$AB = BA$	$A+B = B+A$
Associative	$A(BC) = (AB)C$	$A+(B+C) = (A+B)+C$
Distributive	$A+BC = (A+B)(A+C)$	$A.(B+C) = AB+AC$
Absorption.	$A(A+B) = A$	$A+(AB) = A$
Demorgans Law.	$\overline{AB} = \bar{A} + \bar{B}$	$\overline{A+B} = \bar{A}\bar{B}$



$$(W+X).(W+Y) = W+(X.Y)$$



Property...



$$(\bar{X}+Y) + (\bar{X}\bar{Y})$$

$$0 = A0$$

$$A = A1$$

$$A = AA$$

$$0 = \bar{A}A$$

$$(\bar{X}+Y)(\bar{X}\bar{Y})$$

$$\bar{X}\bar{Y}(\bar{X}+Y)$$

$$\bar{X}\bar{Y}(\bar{X}\bar{Y})$$

$$AB = BA$$

$$A(BA) = (AB)A$$

$$A+B = (A+B)A$$

$$A = (AB) + A$$

$$A = (B+A)A$$

Poynting Vector: $\vec{S}$ wave Propagating in which Direction

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

: Gives the direction of Energy flow.

: Energy transmitted Per Unit time Per Unit area.

: Units: $\frac{\text{Watt}}{\text{m}^2}$

: Same as Intensity.

Magnitude of $\vec{S}$:

$$S = \frac{\vec{E} \cdot \vec{B}}{\mu_0} = \frac{E_0 \sin(\omega t - kx) B_0 \sin(\omega t - kx)}{\mu_0}$$

$$S = \frac{E_0 B_0 \sin^2(\omega t - kx)}{\mu_0}$$

$$i. S_{avg} = \frac{E_0^2}{2\mu_0 c}$$

$$ii. S_{avg} = \frac{B_0^2 c}{2\mu_0}$$

$$S_{avg} = \frac{E_0 B_0}{2\mu_0}$$

$$\frac{E_0}{B_0} = c \quad E_0 = cB_0 \quad B_0 = \frac{E_0}{c}$$

$S = 4\pi \cdot \dots \rightarrow$ Energy density.

Waves:

Radio Waves

γ rays x-rays UV Visible IR Micro Radio

- : Accelerated Motion of charges in wires
- : 500kHz to 1000MHz
- : Radio television.

Microwaves

- : frequency (f): $3 \times 10^8 \text{Hz} < f < 3 \times 10^{11} \text{Hz}$
- : $3 \times 10^8 < f < 3 \times 10^{11} \text{Hz}$
- : Radar systems, Microwave ovens.

iii. Infrared wave:

- : frequency (f): $3 \times 10^{11} \text{Hz} < f < 4 \times 10^{14} \text{Hz}$
- : Hot bodies
- : $3 \times 10^{11} < f < 4 \times 10^{14} \text{Hz}$

iv. Visible waves:

v. IR

vi. Micro

vii. Radio

$$i. B_{\text{rms}} = \frac{B_0}{\sqrt{2}}$$

$$ii. E_{\text{rms}} = \frac{E_0}{\sqrt{2}}$$

$$i. I = \frac{U}{A \Delta t} = \frac{\text{Energy}}{\text{area} \times \text{time}} \quad (\text{Intensity})$$

$$ii. U = U_{\text{avg}} (c \Delta t) A$$

where,

$U \rightarrow$ Energy of electromagnetic wave

$\Delta t \rightarrow$ Time taken to cross.

